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Project Title:	Using Remobilizing Surfactants to Enhance the Thermocapillary Migration of Bubbles Retarded by the Adsorption of Surfactant Impurities		
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Task Description:	This research is a joint collaboration between the Levich Institute and Department of Chemical Engineering at the City College of New York, and the National Center for Microgravity Research at NASA's Glenn Research Center. The focus of the proposed research is to develop methods to enhance the thermocapillary migration of bubbles moving in a continuous liquid phase which have been retarded by the adsorption onto their surfaces of surfactant impurities present in the liquid. We propose to use "remobilizing" surfactant molecules which kinetically rapidly exchange between the bulk and the surface and are at high bulk concentrations. Because the remobilizing surfactant is present at much higher concentrations than the impurity, it adsorbs to the bubble surface much faster than the impurity when the bubble is formed, and thereby prevents the impurity from adsorbing onto the surface. In addition the rapid kinetic exchange and high bulk concentration maintain a saturated surface with a uniform surface concentration. This prevents retarding surface tension gradients and keeps the thermocapillary velocity high. The overall objective of the research program is to increase the thermocapillary velocity of bubbles whose interfacial mobility has been retarded by the adsorption of a surfactant impurity. Impurities adsorb onto the surface of the bubble and are swept to the back end of the bubble creating a cap which rigidifies the surface. To increase the mobility, we will use remobilizing surfactants. These are surfactants which are present at high concentrations in the bulk, and which kinetically exchange rapidly between the bubble surface and liquid subphase adjacent to the surface. Due to the rapid exchange the subsurface and surface are in equilibrium . As a consequence of the high concentration , the surface becomes saturated with a monolayer which is uniform in surface concentration and therefore does not create retarding Marangoni stresses. In addition the high concentration insures that the impurity does not adsorb onto the surface when the bubble is created. The research program is a ground based study and has both experimental and theoretical parts: Specifically, we will construct an apparatus in which bubbles migrate upwards by both buoyancy and thermocapillarity. Two continuous phases will be used: a glycerol/water mixture with high viscosity which results in reduced bubble velocities, and pure water which has greater bubble velocities. The bubble size and the temperature gradient that will be used are such that buoyancy does not mask thermocapillary effects. For the glycerol/water system the measured kinetic exchange rate of polyethylene oxide surfactants should be fast enough to remobilize the surface since the migration velocity is not too large. For water as the continuous phase, the larger velocities require surfactants with kinetic exchange rates greater than the polyethoxylates. We will concentrate our efforts on lower chain alcohols, where reports in the literature show that the kinetic exchange may be sufficiently fast, and sugar surfactants whose kinetic exchange is unknown (and needs to be measured) but whose solubilities are very large and thereby easily satisfy the requirement of high bulk concentration. The kinetic exchanges will be determined by measuring dynamic tension reduction using the pendant bubble technique as we have done previously for the polyethoxylate surfactants. For the water system, we will first undertake buoyancy only experiments to identify the remobilizing surfactant. Theoretically, we will redevelop the theory to incorporate micellization, since at high concentrations micelles form in the bulk and equilibration of the surface by exchange with micelles can effectively remobilize the surface. We will develop an asymptotic theory for remobilization valid at high bulk concentration and Peclet number. It is well recognized that microgravity processes must rely on mechanisms other than buoyancy to move bubbles from one region to another in a continuous liquid phase. One suggested method is thermocapillary migration in which a temperature gradient is applied to the continuous phase. When a fluid particle contacts this gradient, one pole of the particle becomes warmer than the opposing pole. The interfacial tension between the bubble phase and the continuous phase usually decreases with temperature. Thus the cooler pole is of higher interfacial tension than the warmer pole, and the interface is tugged in the direction of the cooler end. This thermocapillary or thermally induced Marangoni surface stress causes a fluid streaming in the continuous phase from which develops a viscous shear traction and pressure gradient which together propel the particle in the direction of the warmer fluid. A significant and as yet unresolved impediment to the use of thermocapillary migration to direct bubble or drop motion is that these migrations can be significantly retarded by the adsorption onto the bubble surface of surface active impurities dissolved in the continuous phase. Adsorbed surfactant on the bubble surface is convected by the surface flow from the front to the particle's trailing end forming a rigid cap which decreases the interfacial mobility and thermocapillary velocity. What we hope to accomplish in this research is to demonstrate that thermocapillary migration velocities can be increased by dissolving remobilizing surfactants in the continuous phase. These adsorb onto the surface and form a protective monolayer of uniform surface concentration which maintains a large thermocapillary migration velocity.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Optimizing Mass Transfer in Interfacial Mass Transfer Processes
	During the year 2004, we began experiments to verify remobilization above the critical micelle concentration (CMC) using buoyancy-induced motion. (Thermocapillary migrations will be studied during the next year.) An apparatus was constructed to measure by video tracking the terminal velocities of bubbles rising in a continuous liquid phase in a column, and velocity measurements were undertaken for the clean system and for two nonionic polyethylene oxide surfactants (C12E6; CH3 (CH2)(11)(OCH2CH2) 6OH) and (C10E8; CH3 (CH2)(9)(OCH2CH2)8OH) as a function of the surfactant bulk concentration. Experiments were undertaken with relatively large bubbles (of the order of one millimeter in diameter); to insure that these bubbles remain spherical (so as to match the measurements to theory), a very viscous solution of glycerol and water (70:30 by volume) was used so that velocities remained small enough so that inertial effects did not distort the bubble shape from a spherical form. For the polyethylene oxide surfactants and the air/glycerol/water interface, we undertook, using the pendant bubble apparatus, measurements of the equilibrium surface tension as a function of bulk concentration to obtain the CMC and equilibrium adsorption parameters, and the relaxation in tension as surfactant adsorbs onto an initially clean air-glycerol/water interface to obtain the kinetic constants and the diffusion coefficient of the monomer in the solution. For C12E6 the interfacial tension measurements indicated that for concentrations of surfactant up to the CMC, the diffusive transport rate is slow relative to the rate of surface convection of surfactant, and therefore the bubble interface should not remobilize for concentrations less than the CMC. This was observed. For this surfactant, only lower bounds on the kinetic coefficients were obtained from the dynamic tension measurements. The numbers for the lower bounds suggested that the kinetic rates are also much smaller than the surface convection rate. Since these numbers are lower bounds, the kinetic rates could be faster. However, above the CMC, we observed from the terminal velocity measurements that the interface of the bubbles remains completely immobile, pointing to the fact that even if micelles

Task Progress:	can maintain a uniform monomer concentration, the kinetic limitation on the mass transfers results in the large gradients in tension necessary to maintain an immobile interface. In the case of C10E8, the shorter chain length increases the solubility (the CMC is raised) allowing at the higher concentrations a faster diffusive transport of monomer. In addition the smaller chain length results in less cohesion in the adsorbed layer, and a larger effective adsorption rate constant. The interfacial tension measurements showed an order of magnitude higher critical micelle concentration and effective desorption rate constant. Bubbles rising in a continuous phase containing dissolved C10E8 were observed to first form a tangentially immobile layer at low bulk concentrations of surfactant, but at higher concentrations to partially remobilize. The remobilization began for concentrations just below the CMC, but dramatically increased for concentrations above the CMC indicating the remobilizing effect of micellar exchange. Additional theoretical progress was also made on the asymptotic solution for the terminal velocity for the case below the critical micelle concentration in which surfactant transport between the bulk liquid and the surface takes place in a thin boundary layer next to the surface of the bubble. Solutions earlier were obtained for the case in which bubble velocities were large enough that the fluid inertia was large relative to viscosity (high Reynolds number); new solutions were obtained for the low velocity limit dominated by viscosity. The focus of our research for the final year (2005) of the grant will be threefold: First the remobilization results obtained above the CMC for C10E8 will be compared with the theory we have developed earlier. In this theory, the kinetic exchange of surfactant between the surface and the sublayer of liquid adjoining the surface is assumed to be much faster than the bulk diffusion so that the surface and sublayer are in quasi-equilibrium. In addition, the exchange of surfactant monomer between the bulk and micelles dispersed in the continuous phase is assumed to be much faster than diffusion so that micelles provide a reservoir to maintain the monomer concentration at the CMC. Comparisons of the predicted terminal velocities with experimental measurements will determine whether the theory needs to be extended to include the kinetic effects. Second, remobilization below the CMC will be studied experimentally. To remobilize an interface below the CMC, the surfactant must have a very large solubility so that the diffusive transport rate of monomer to the surface is much faster than the rate of surface convection. Medium chain length alcohols or polyethylene oxide surfactants with a small number of methylene groups relative to the number of ethoxylate groups can have elevated solubilities in water (or polar solvents like the glycerol/water mixture used in the above experiments), and at their solubility limit, may form an insoluble phase of the neat liquid rather than a dispersed micellar phase. We plan to study hexanol as a remobilizing agent. Studies in the literature note that this alcohol retains a very high solubility in water sufficient to insure that the diffusion will be faster than the surface convection rate and demonstrate that micelles do not form at the point of insolubility. In addition, dynamic tension studies of hexanol adsorbing at an air/water interface show that the kinetic rates of exchange of the alcohol between the sublayer and surface are of the same order or larger than the surface convection rate. Hence we should be able to remobilize the interface of rising air bubbles, and we will attempt this remobilization in both aqueous and glycerol/water continuous phases. These experiments will also be compared to the predictions of the boundary layer analysis of the bubble retardation.

Bibliography Type:	Description: (Last Updated: 12/02/2020)
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Articles in Peer-reviewed Journals	Song, Q.; Couzis, A.; Somasundaran, P.; Maldarelli,C. "Theory and Supporting Experiments on the Adsorption of Surfactant From a Micellar Solution to a Clean AIr/Water Interface" Journal of Colloid and Interface Science (submitted) , Jul-2004
Articles in Peer-reviewed Journals	Palaparthi, R.; Papageorgiou, D.; Balasubramanian, R.; Maldarelli, C. "Remobilization of Moving Bubbles Retarded By Surfactant Adsorption at Concentrations Above the CMC" Physical Review Letters (submitted) , Jul-2004
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Presentation	Taneja, A.; Subramanian, B., Papageorgiou, D., Maldarelli, C. "Surfactant Retardation of Bubble Motion for Concentrations Above the CMC" AIChE National Meeting Nov-2003